

Thermodynamics

Lecture 5B:

- Energy analysis of control volumes

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Learning objectives (I)

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. *Comprehension:* Describe the first and second laws in the student's own language.**
- III. *Application:* Solve simple single-answer problems using the first law.**
- IV. *Analysis:* Solve problems requiring both the first and second laws.
- V. *Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

Learning objectives (II)

- VI. Evaluation: Determine and describe second law fallacies in proposed power cycles.
- VII. Evaluation: Judge when classical thermodynamics is not the appropriate analysis tool.
- VIII. Evaluation: Find and correct errors in the student's own solutions and in those of others.
- IX. *Analysis and Evaluation:* Search appropriate databases and the literature to find required thermodynamic data, and if the data are not available the student can select appropriate procedures and predict the values of the data.**
- X. Knowledge: Briefly outline the history of the field of thermodynamics.

After this lecture

- Apply the 1st law of thermodynamics to steady-flow systems.
- Solve energy balance problems for common steady-flow devices such as turbines, nozzles, compressors, valves, mixing chambers and heat exchangers.

Recap Lecture 5A

Energy analysis of control volumes

Conservation of mass

$$\sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i} = \frac{dm_{CV}}{dt}$$

$$\frac{\dot{E}_{mass,in}}{\dot{m}_{in}} = e = u + \frac{V^2}{2} + gz$$

1st law for control volumes: to be continued

$$\dot{W}_{tot} = \dot{W}_s - \dot{W}_{flow,in} + \dot{W}_{flow,out}$$

$$\frac{dE_{system}}{dt} = \dot{Q} - \dot{W}_{tot} + \left(\sum_{in}^n \dot{E}_{mass,in} - \sum_{out}^n \dot{E}_{mass,out} \right)$$

Total energy of a flowing fluid

Include the flow work/energy into the total energy transport by mass.

Non-flowing fluid $e = u + ke + pe = u + \frac{V^2}{2} + gz$

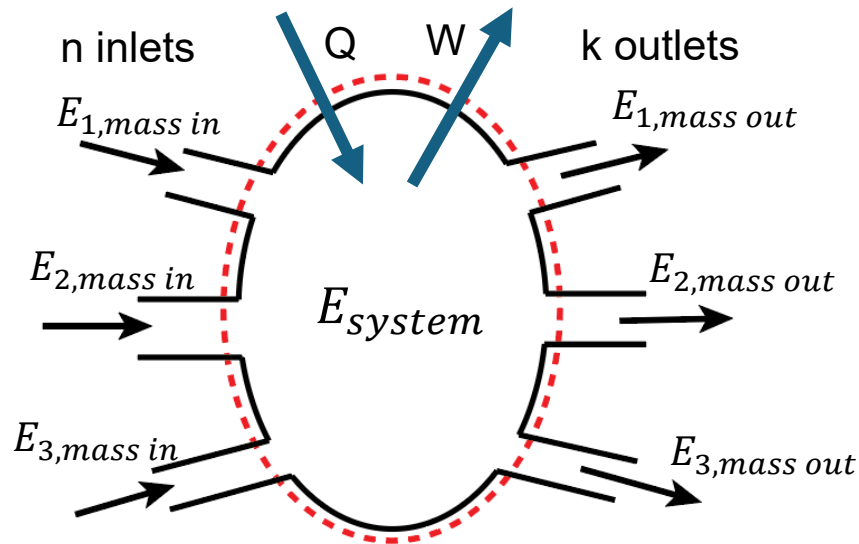
Flowing fluid $\theta = e + pv = u + ke + pe + pv$
 $\theta = h + \frac{V^2}{2} + gz$

Energy transport by mass $E_{mass} = m\theta = m\left(h + \frac{V^2}{2} + gz\right)$

Rate of energy transport by mass $\dot{E}_{mass} = \dot{m}\theta = \dot{m}\left(h + \frac{V^2}{2} + gz\right)$

Energy balance of a control volume

$$\frac{dE_{system}}{dt} = \dot{E}_{in} - \dot{E}_{out}$$



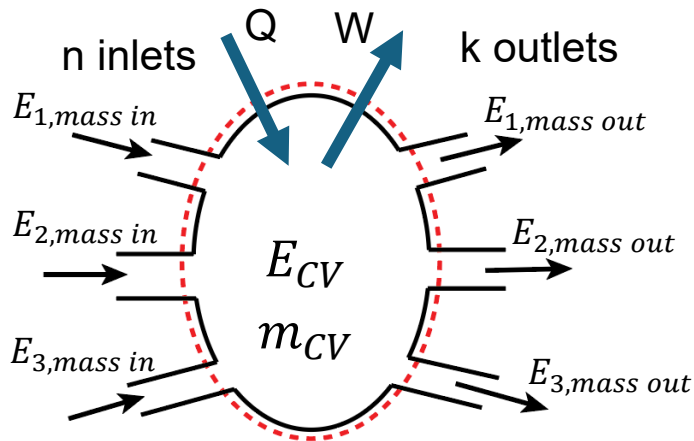
Steady flow systems (I)

- Steady flow systems

$$\sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i} = \frac{dm_{CV}}{dt}$$

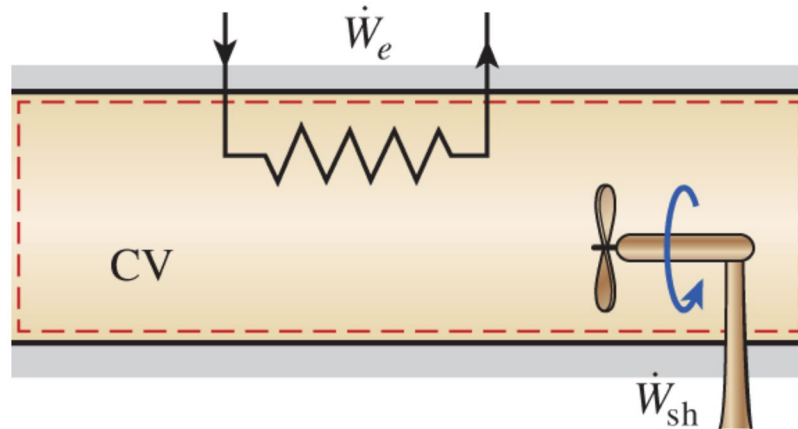
$$\dot{E}_{in} - \dot{E}_{out} = \frac{dE_{CV}}{dt}$$

$$\dot{Q} - \dot{W}_s + \sum_{i=1}^n \dot{m}_{in,i} \left(h_{in,i} + \frac{V_{in,i}^2}{2} + gz_{in,i} \right) - \sum_{i=1}^k \dot{m}_{ex,i} \left(h_{ex,i} + \frac{V_{ex,i}^2}{2} + gz_{ex,i} \right)$$

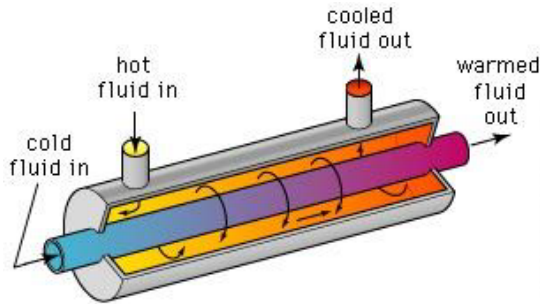


Steady flow systems (II)

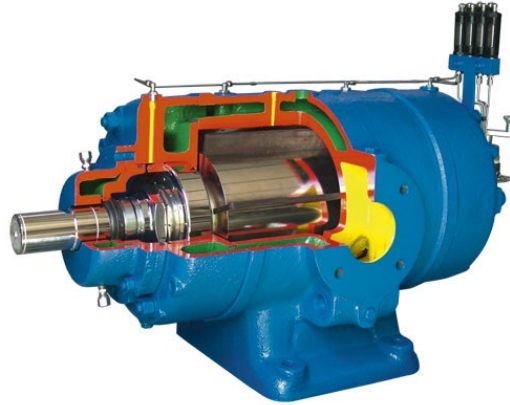
- Control volume is typically constant, so there is no boundary work.
- Work required to push in/out fluid is taken care of by enthalpy.
- Shaft work and electrical work are typically the only types of work involved in CVs.



Examples of steady flow devices



Heat exchangers



Air compressor



Nozzle

Image credit: Wikipedia, Britannica

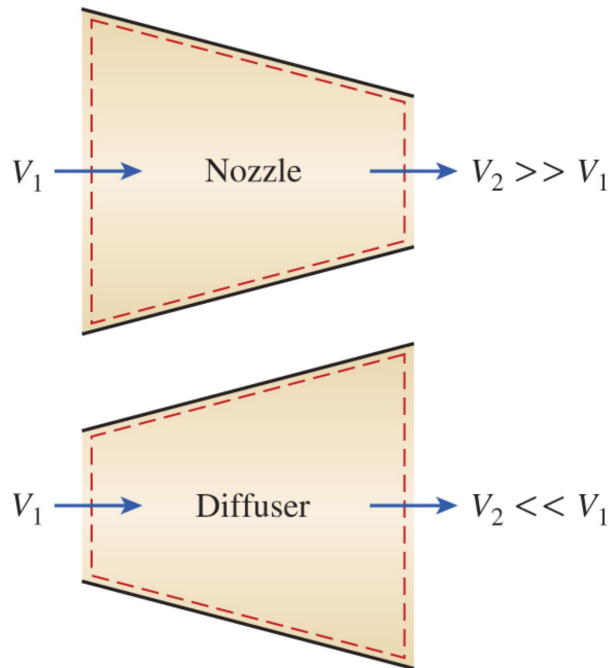
Nozzles & Diffusers

- Increase/decreasing fluid velocity by decreasing/increasing fluid pressure.

$$\dot{m}_{in} = \dot{m}_{ex}$$

Assumed adiabatic

$$\dot{Q} = 0; \quad \dot{W}_s = 0; \quad \Delta PE = 0$$



Compressors: Example in groups of two, 10 min

Given:

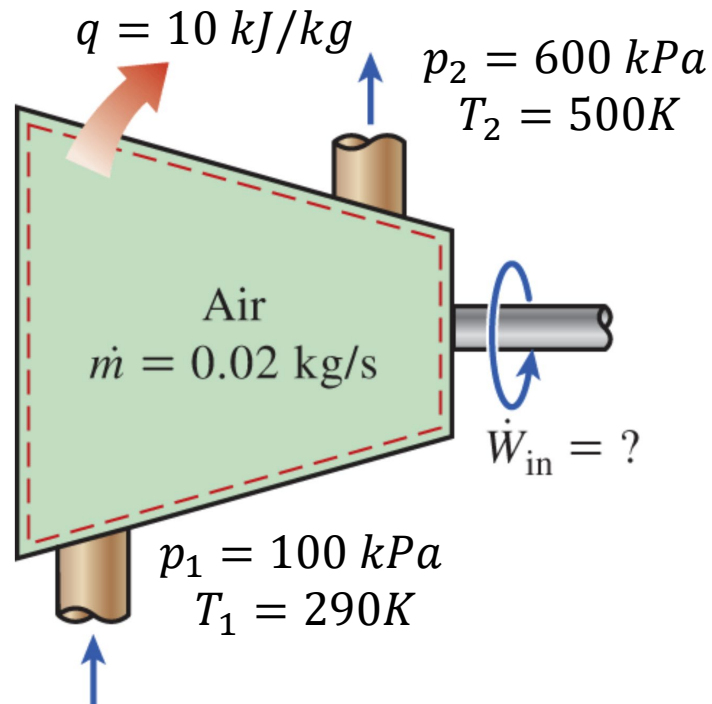
A compressor operates at steady state and compresses air (ideal gas) from state 1 to state 2 as shown in the sketch.

Determine:

1. P-V diagram of the process.
2. The required shaft work.

To Do:

- Go to pollev.com/claudiohail
- Enter the first names of your group members, e.g. Olivia and Liam.
- Submit your answers to 1 and 2.



Ideal gas properties of air (Table A-17)

T K	h kJ/kg	P_r	u kJ/kg	v_r
250	250.05	0.7329	178.28	979.0
260	260.09	0.8405	185.45	887.8
270	270.11	0.9590	192.60	808.0
280	280.13	1.0889	199.75	738.0
285	285.14	1.1584	203.33	706.1
290	290.16	1.2311	206.91	676.1
295	295.17	1.3068	210.49	647.9
298	298.18	1.3543	212.64	631.9
300	300.19	1.3860	214.07	621.2

T K	h kJ/kg	P_r	u kJ/kg	v_r
470	472.24	6.742	337.32	200.1
480	482.49	7.268	344.70	189.5
490	492.74	7.824	352.08	179.7
500	503.02	8.411	359.49	170.6
510	513.32	9.031	366.92	162.1
520	523.63	9.684	374.36	154.1
530	533.98	10.37	381.84	146.7
540	544.35	11.10	389.34	139.7
550	554.74	11.86	396.86	133.1

Compressors: Example solution

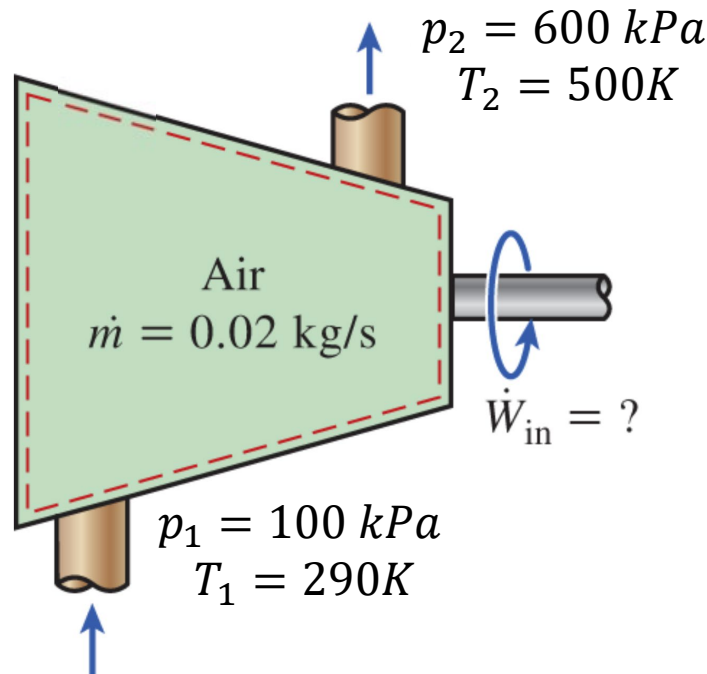
Compressors: Example

- Compress a fluid to a higher pressure by supplying shaft work.

$$\dot{m}_{in} = \dot{m}_{ex}$$

Assumed adiabatic

$$\dot{Q} = 0; \dot{W}_s \neq 0; \Delta ke = 0; \Delta pe = 0;$$

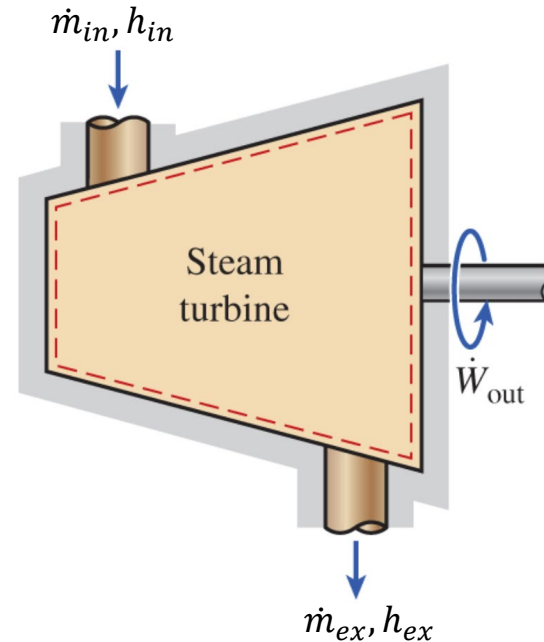


Turbine

- Extract useful shaft work by expanding a fluid to a lower pressure.

$$\dot{m}_{in} = \dot{m}_{ex}$$

Assumed adiabatic

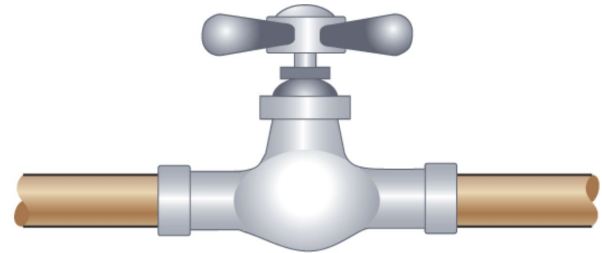


Throttling valves

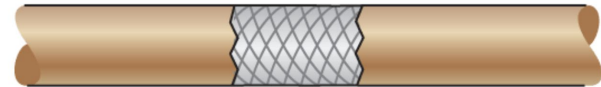
- Device that causes a large pressure drop in the fluid.

$$\dot{m}_{in} = \dot{m}_{ex}$$

For an ideal gas:



(a) An adjustable valve



(b) A porous plug



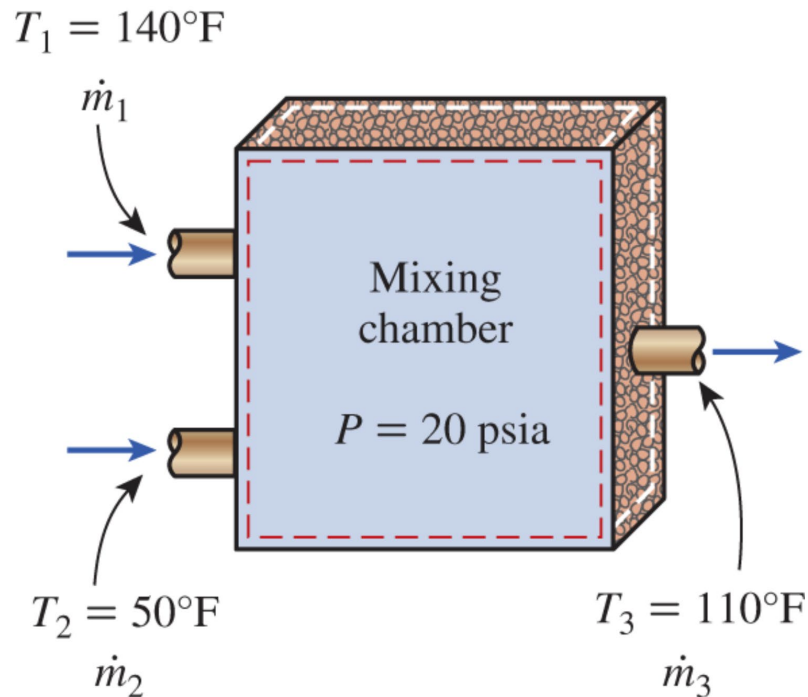
(c) A capillary tube

Mixing chamber

A device that combines multiple streams of fluids.

$$0 = \sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i}$$

$$\dot{Q} = 0; \dot{W}_s = 0; \Delta ke = 0; \Delta pe = 0;$$

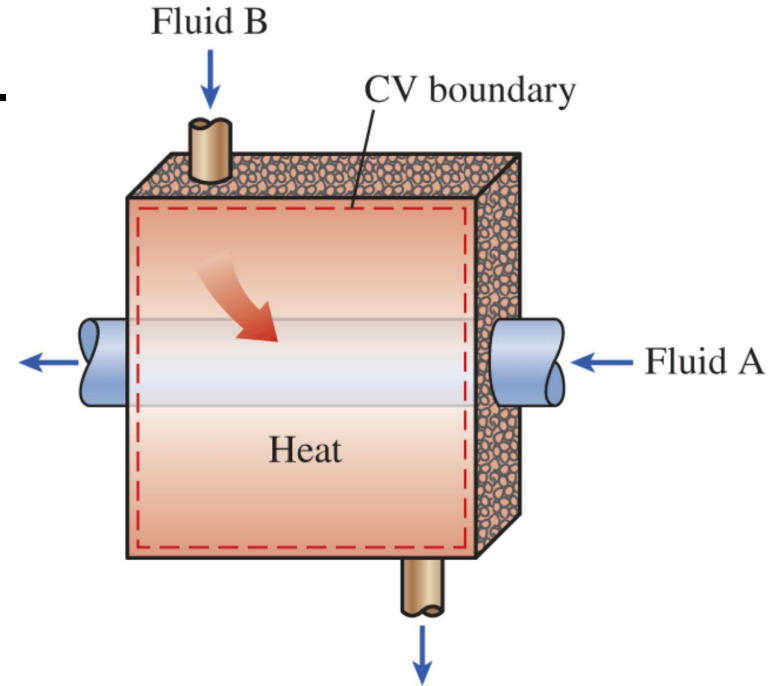


Heat exchanger

A device where fluid streams exchange heat without mixing.

Each fluid
$$0 = \sum_{i=1}^n \dot{m}_{in,i} - \sum_{i=1}^k \dot{m}_{ex,i}$$

$$\dot{Q} = 0; \dot{W}_s = 0; \Delta ke = 0; \Delta pe = 0;$$



Pipe or duct flow

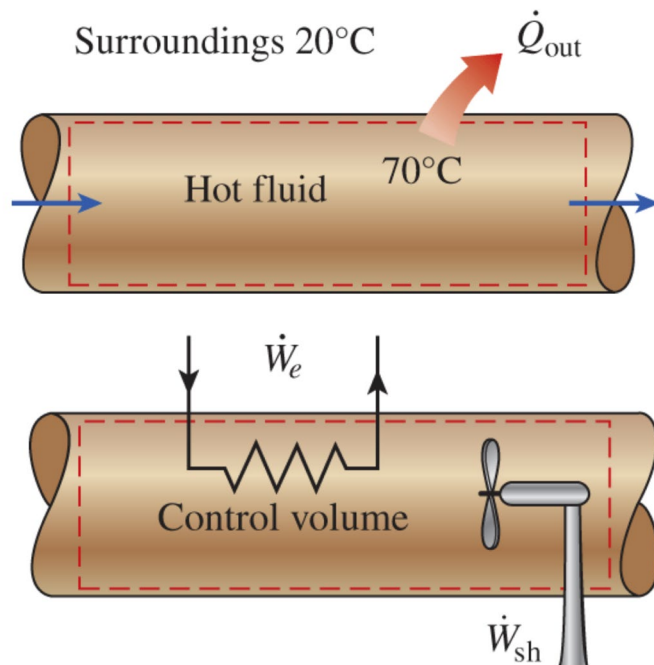
Any general device for the transport of fluid accompanied by heating/cooling, or electrical/shaft work.

$$\dot{m}_{in} = \dot{m}_{ex}$$

In general:

$$\dot{Q} \neq 0; \dot{W}_s \neq 0; \Delta ke = 0; \Delta pe = 0;$$

$$0 = -\dot{W}_s + \dot{m}_{in}h_{in} - \dot{m}_{ex}h_{ex}$$



Module 1 Review Topics

Fill out this survey for
suggest a topic and HW
problem for the Module 1
Review during the lecture
Wednesday 9/23/26.



Summary

- Apply the 1st law of thermodynamics to steady-flow systems.
- Solve energy balance problems for common steady-flow devices such as turbines, nozzles, compressors, valves, mixing chambers and heat exchangers.

Next up

- Apply the 1st law of thermodynamics to unsteady-flow processes.
- Review concepts of module: pages 1-5.